32-bit microcontrollers

HC32L072_HC32L073_HC32F072 series MCU development tools

User manual

Rev2.3 December 2023

Applicable objects

产品系列	产品型号	产品系列	产品型号
L 系列	HC32L072 HC32L073	**F 系列**	HC32F072

本手册以 HC32L073PATA 为例进行说明。

Statement

- ★ Xiaohua Semiconductor Co., Ltd. (hereinafter referred to as "XHSC") reserves the right to change, correct, enhance, and modify Xiaohua Semiconductor products and/or this document at any time without prior notice. Users can obtain the latest relevant information before placing an order. XHSC products are sold in accordance with the sales terms and conditions stated in the basic purchase and sales contract.

- ★ Customers should select appropriate XHSC products for your application and design, verify and test your application to ensure that your application meets the corresponding standards and any safety, security or other requirements. Customers shall bear all responsibility for this.

- ★ XHSC hereby confirms that no intellectual property license is granted, either expressly or impliedly.

- ★ Resale of XHSC products, if its terms are different from those specified herein, will invalidate any warranty commitment of XHSC for such products.

- ★ Any graphics or words with "®" or "™" logos are trademarks of XHSC. All other product or service names displayed on XHSC products are the property of their respective owners.

- ★ The information in this notice supersedes and replaces the information in previous versions.

Table of contents

1 Overview

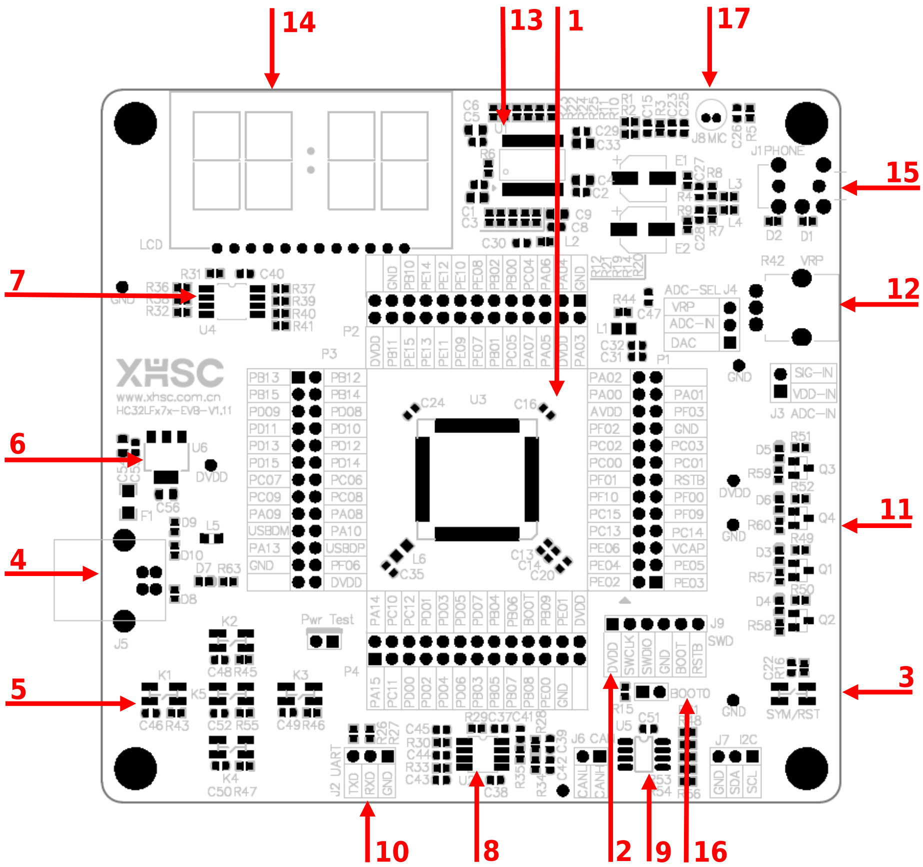
1.1 Introduction to development tools

This series of Evaluation Board (EVB) is a development tool designed based on the HC32L073PATA chip. This article introduces how to use the EVB. It mainly describes the hardware resources used by the chip, the software development environment, the installation and use instructions of the development environment, debugging methods, board codes, and tool usage, etc., aiming to help developers carry out development work conveniently.

For chip specifications, please refer to the corresponding "Data Sheet".

1.2 Introduction to circuit board components

MCU minimum s ystem
1, HC32L07/HC32F07
2, SWD/ISP debugging programming port
3, Reset button (RST)
4, USB-FS interface
5, User buttons (K1~K5)
6, LDO-3V3
7, SPI FLASH
8, CAN PHY
9, I2C EEPROM/IF
10 , UART IF
11 , LED (4)
12, AD/DA IF
13, CODEC
14, LCD display
15, Audio IF
16, Mode selection (BOOT0)
17, MIC microphone



Note 1:

Please read the instruction card inside the box before use.

MCU minimum system and peripherals			
1	HC32L07/ HC32F07	11	LED（4)
2	SWD/ISP debugging programming port	12	AD/ DA IF
3	Reset button (RST)	13	CODEC
4	USB-FS interface	14	Liquid crystal display (LCD)
5	User buttons (K1~K5)	15	Audio IF
6	LDO	16	Mode selection (BOOT0)
7	SPI FLASH	17	MIC microphone
8	CAN PHY	-	-
9	I2C EEPROM/ IF	-	-
10	UART IF	-	-

2 Hardware circuit

2.1 Circuit specifications

MCU supports wide voltage range (1.8-5.5V) and wide temperature range (-40-105℃).

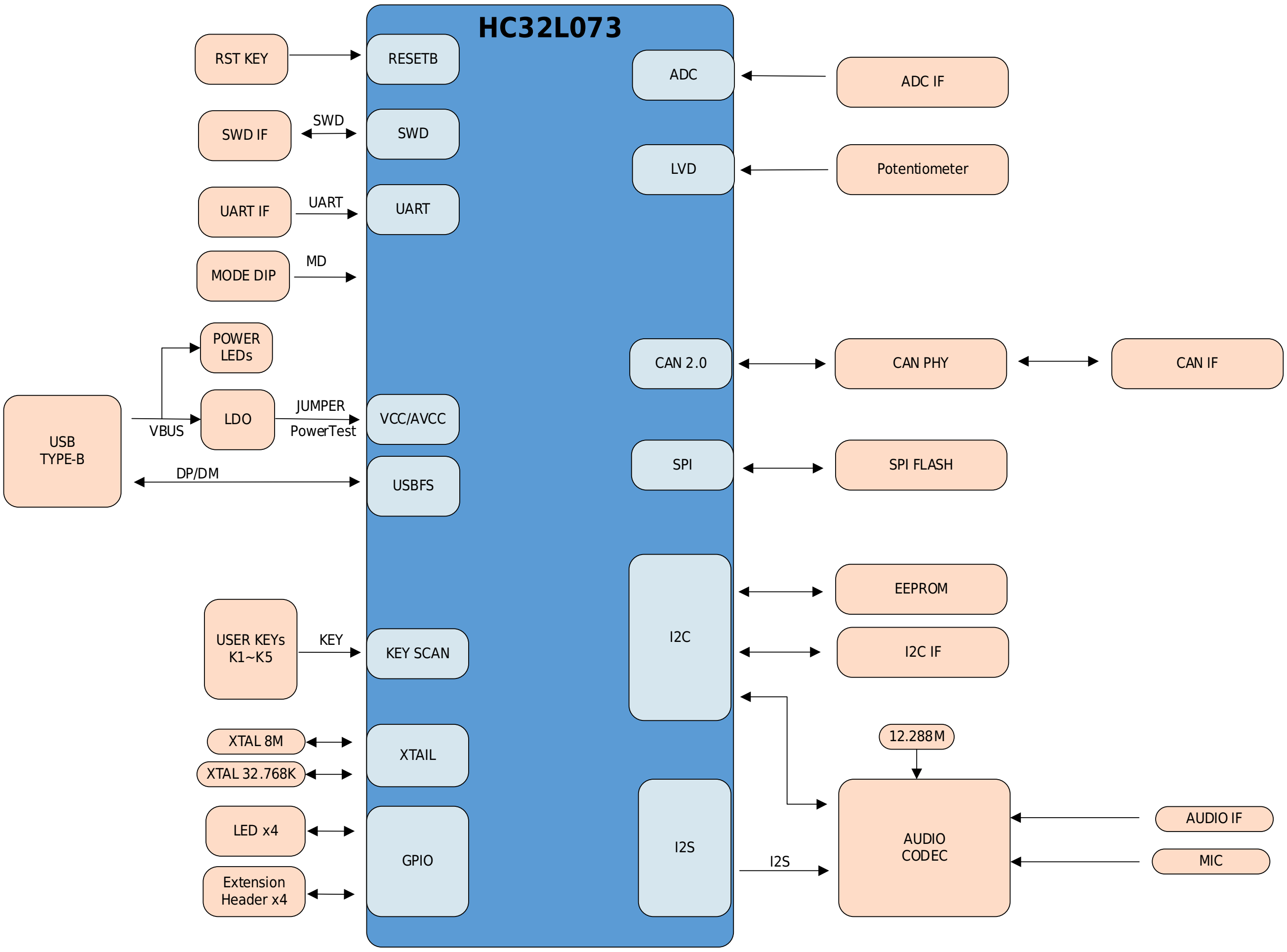
Due to the limitation of LCD screen on the board, it is recommended that the operating temperature of the development tool is -40℃~80℃, and the MCU operating voltage is 3.3V.
During use, please ensure that the operating conditions do not exceed the absolute maximum ratings.

2.2 Hardware description

It is recommended to go to Xiaohua Semiconductor's official website https://www.xhsc.com.cn to find the corresponding chip model and download it.

2.2.1 System overview

The EVB hardware system is shown in the figure below.



2.2.2 USB interface

USB function and EVB power supply, please ensure that the USB host has sufficient power supply capacity.

2.2.3 Debug interface

EVB configures SWD interface, users use this interface for debugging.

2.2.4 UART interface

EVB configures J2 interface, users use this interface for serial port debugging.

2.2.5 Buttons

EVB provides 6 physical buttons for users in the MCU minimum system area:

Designator	Pin/Function
SYM/ RST	RESET/Reset button
K1	PC12/User button
K2	PD03/User button
K3	PD02/User button
K4	PD04/User button
K5	PD06/User button

2.2.6 Light indicators

EVB is equipped with 6 indicators, namely power indicator and user indicator.

Designator	Pin/Function
PWR	MCU minimum system power indicator
D5	PE03/Red user indicator
D6	PE02/Yellow user indicator
D3	PE01/Blue user indicator
D4	PE00/Green user indicator

2.2.7 Clock

EVB is equipped with 2 sets of external clocks, namely low-speed clock Y2 and high-speed clock Y3.

Designator	Pin/Function	Connected peripherals
Y2	PC14/ XTLI	32.768KHz crystal
	PC15/ XTLO	
Y3	PF00/ XTHI	High-speed crystal oscillator (subject to actual board-level use)
	PF01/ XTHO	

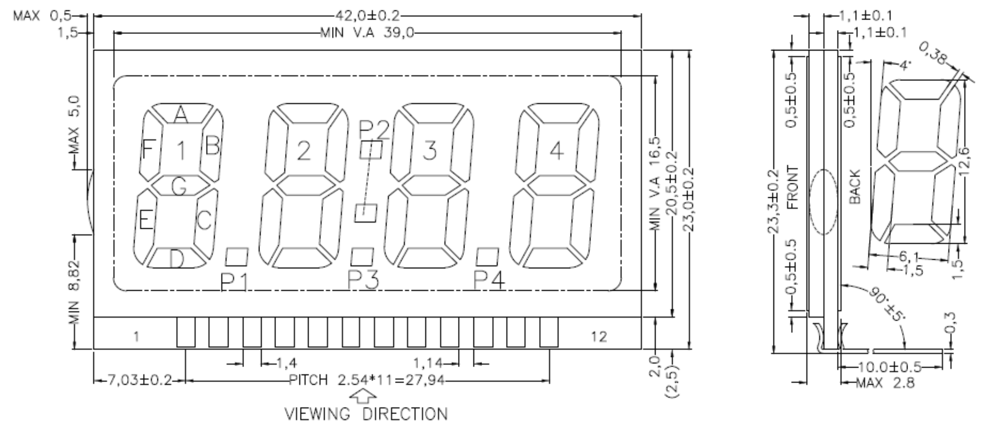
2.2.8 Test pins

EVB is equipped with 4 sets of 2x13 test pin headers, connected to MCU pins, to provide user testing or expansion functions.

2.2.9 LCD

LCD drive mode: 1/4Duty, 1/3Bias, working voltage 3V3.

For other information, please refer to the figure below:



PIN	1	2	3	4	5	6	7	8	9	10	11	12
COM1	COM1				1F	1A	2F	2A	3F	3A	4F	4A
COM2		COM2			1G	1B	2G	2B	3G	3B	4G	4B
COM3			COM3		1E	1C	2E	2C	3E	3C	4E	4C
COM4				COM4	P1	1D	P2	2D	P3	3D	P4	4D

The default configuration of the EVB hardware is the external capacitor voltage division mode for LCD Bias voltage. If you need other modes, please adjust according to the data sheet.

2.2.10 Jumper setting

There are three sets of jumper pins J2/ BOOT0/ PwrTest on the EVB. The jumper pin status needs to be confirmed before powering on. The specific settings are as follows:

Designator	Function	Setup	Default
PwrTest	MCU power consumption test selection	Closed: Normal working mode	Closed
		Opened: Connect a multimeter in series to test MCU power consumption	
BOOT0	MCU mode selection	Opened: User mode	Opened
		Closed: BOOT mode	
J3	ADC detection selection input source	Closed: Use AVDD as a detection signal	Closed
		Opened: Connect external signal to SIG-IN as a detection signal	

3 Driver library

This series of chips supports third-party IDE development, mainly supporting mainstream development environments such as IAR and Keil MDK. Please refer to the "Xiaohua Semiconductor MCU Development Environment Usage" document to familiarize yourself with the relevant configuration and usage.

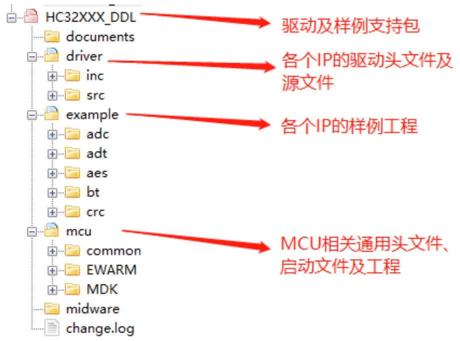
After familiarizing yourself with the IDE development tools, please go to Xiaohua Semiconductor's official website https://www.xhsc.com.cn to find the corresponding chip model HC32L073PATA and download the driver library and samples.

3.1 DDL Version

Please confirm that you have obtained the latest version of the driver library and samples from the official website.

3.2 HC32L073_DDL

The main structure examples of the driver library and sample support package can be found in the figure below (the specific structure is subject to the DDL support package actually used).



driver:

This directory mainly includes the APIs used by each IP operation, header files and source files of the data structure. Users can directly use them in their own applications, or use them to familiarize themselves with the operation of the underlying registers.

example:

This directory mainly includes the usage routines of the common functions of each IP (supporting both IAR and Keil development tools). Users can use this example to quickly familiarize themselves with the implementation methods of the common functions of each IP and the use of the driver library. This example can be directly downloaded, debugged and run with the STK supporting this series of chips.

mcu:

This directory mainly includes the basic header files and startup files required for this series of MCU projects, as well as IAR and Keil project files and their configuration files.

3.3 HC32L073_template

The template mainly provides the system minimum project corresponding to this series of MCUs. If users want to develop their own applications (including drivers for special needs) for specific chip models, they do not need to build projects from scratch. They can directly use this template to directly develop application-related drivers or applications.

3.4 IDE support package

The IDE support package provides the pack file for the chip used in Keil MDK.

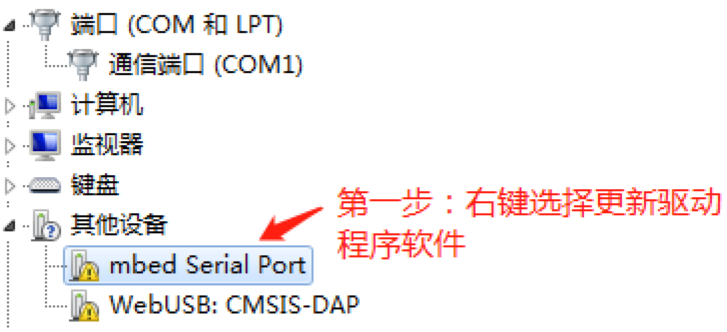
Note:

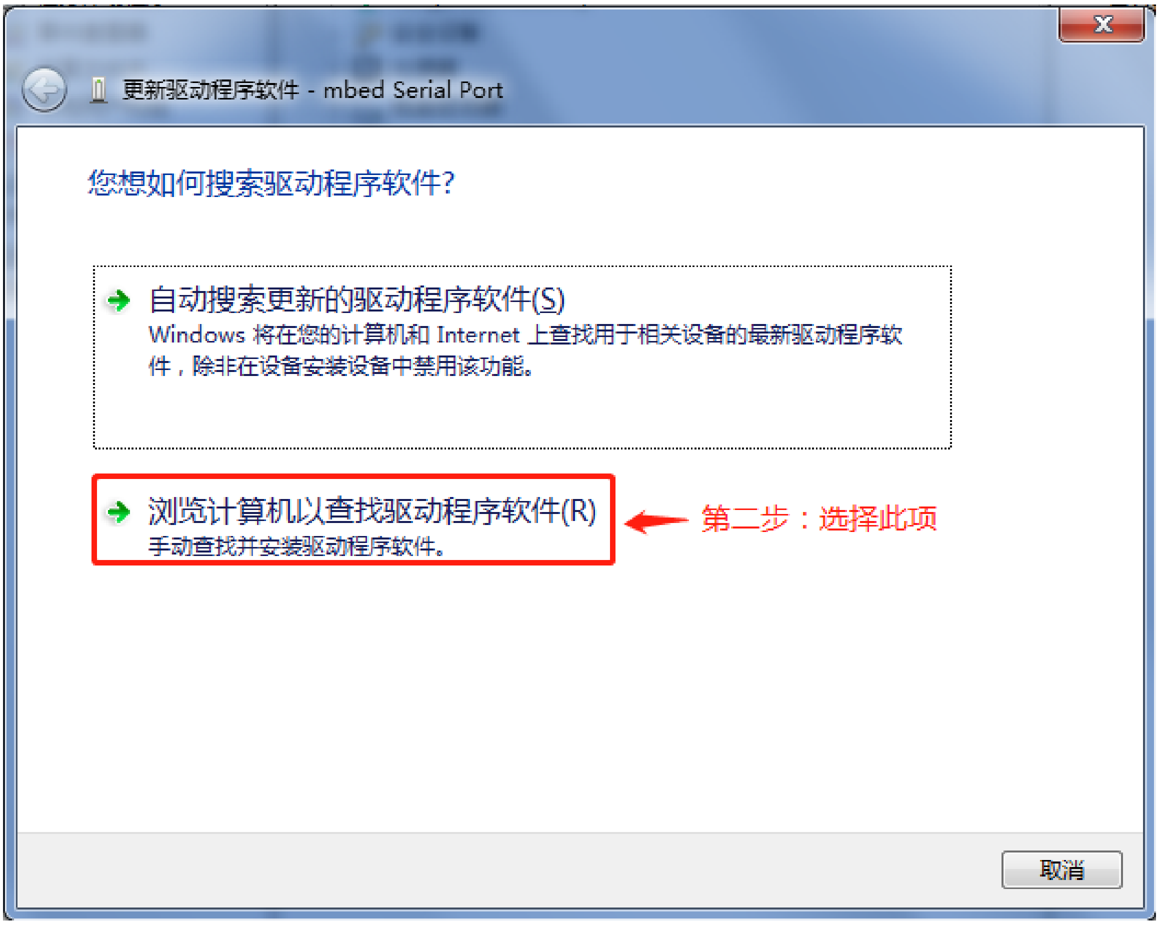
*When using Keil as a development tool for debugging and downloading, you need to ensure that the Keil tool support package for this series of chips is correctly installed, or copy the *.FLM file in the directory ~/mcu/MDK/ to the Keil installation path (~/Keil/ARM/Flash/) of the personal computer, and configure and select the *.FLM file suitable for the chip you are using in the Keil project configuration download option.*

4 Tool use

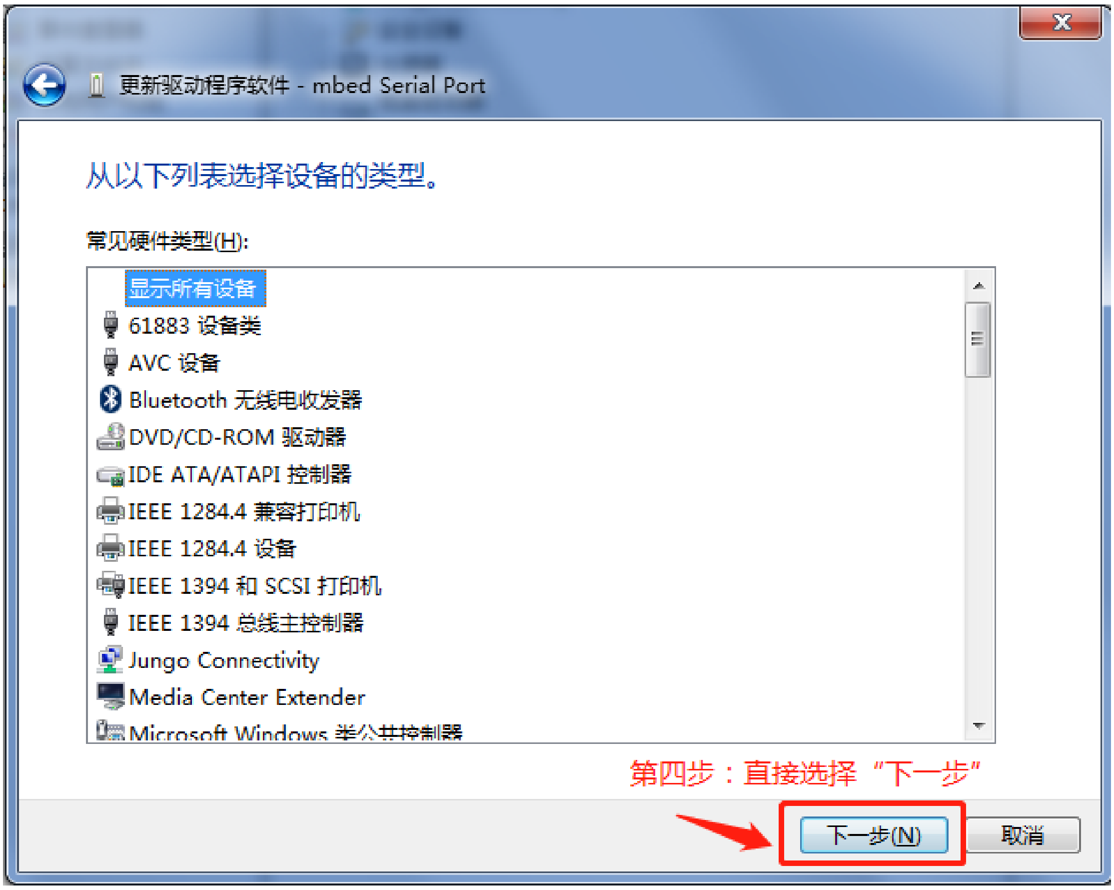
4.1 Debug description

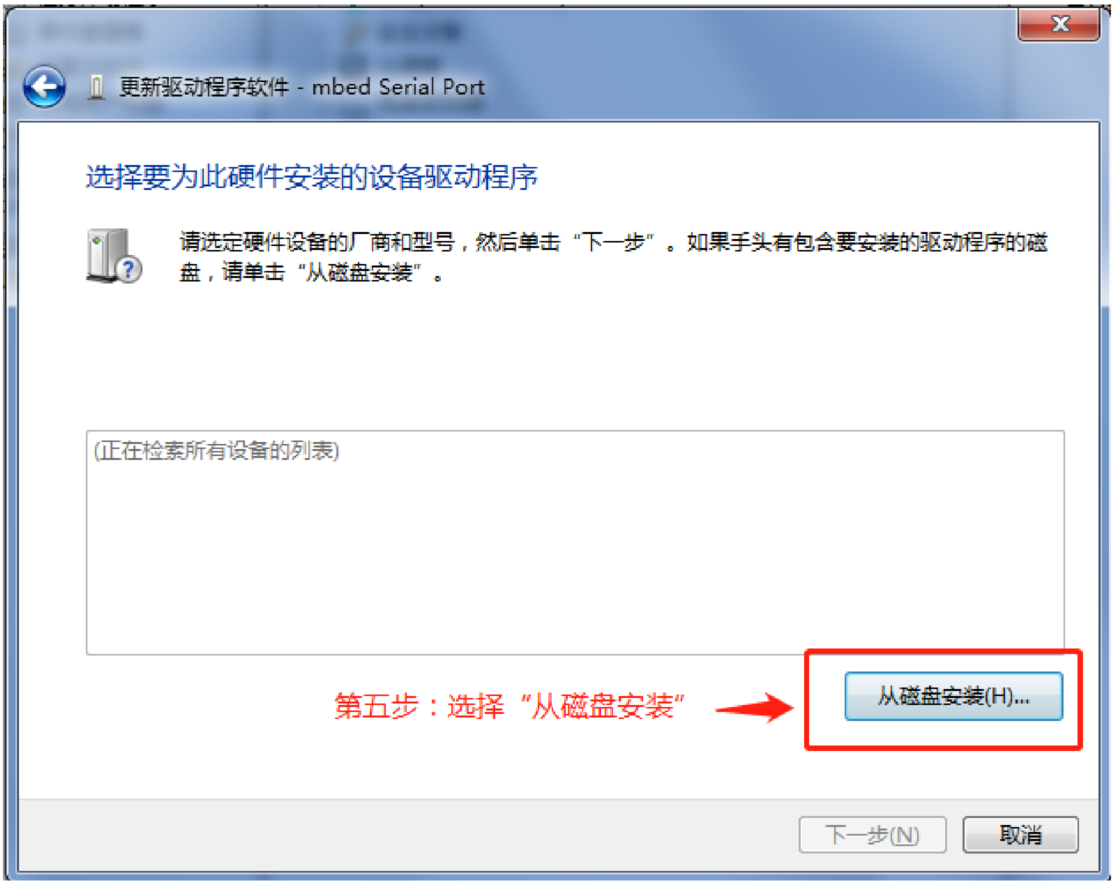
CMSIS DAP supports PCs running WIN10 and above. If you are using Windows 7, please ask the agent or FAE for the driver file. After opening the device manager, follow the steps below to install:

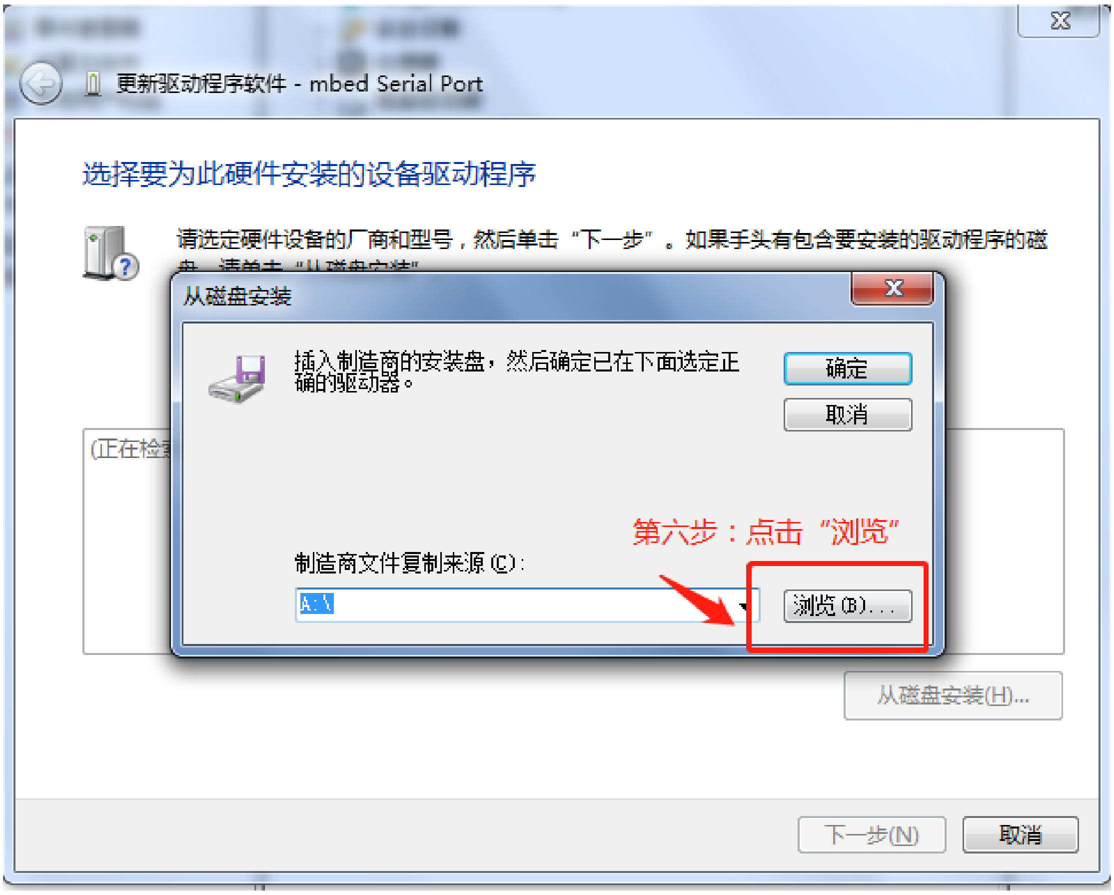


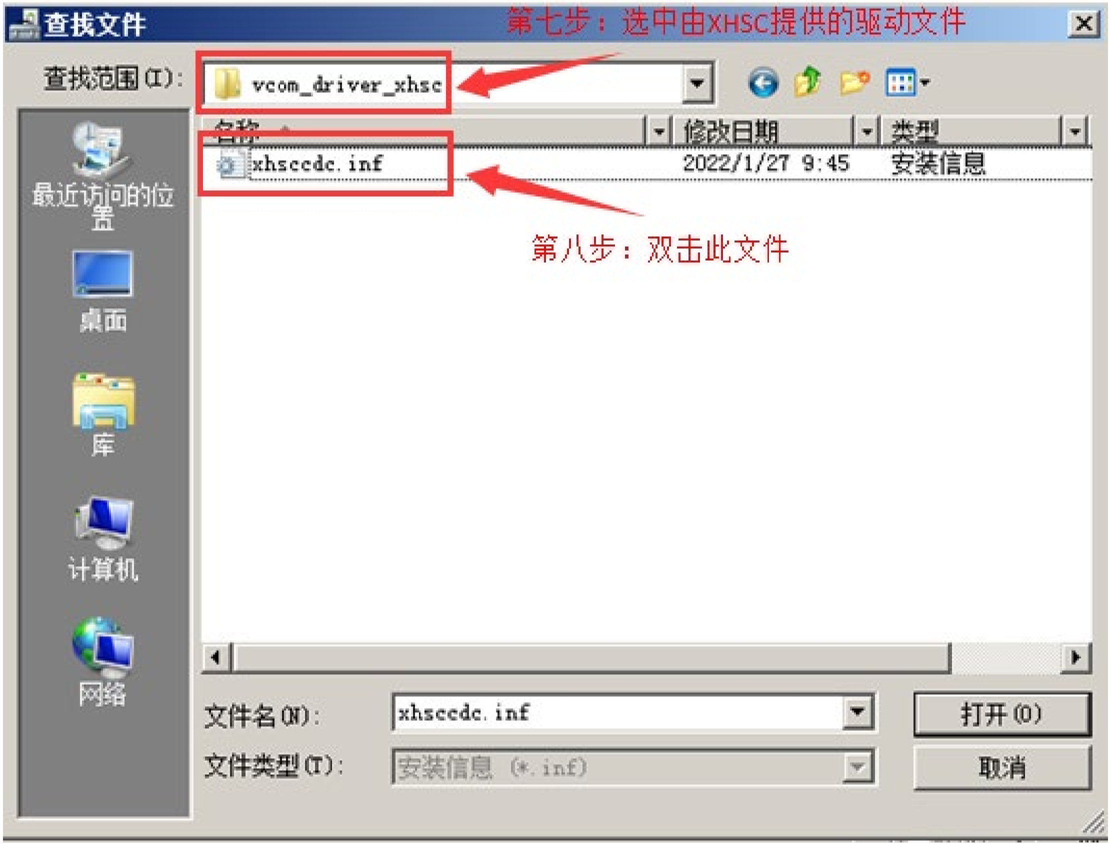


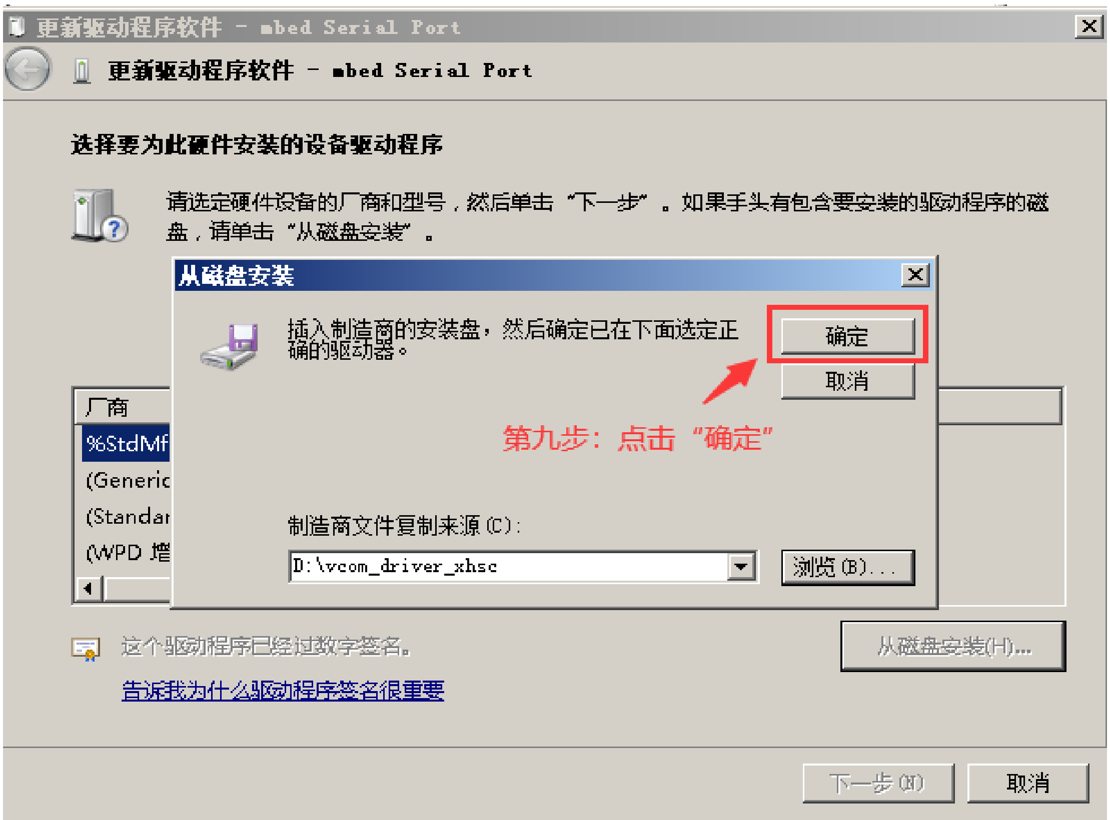

更新驱动程序软件 - mbed Serial Port
浏览计算机上的驱动程序
在以下位置搜索驱动程序:
浏览(R)...
包括子文件夹(I)
→ 让我从计算机上的可用驱动程序列表中选取(L)
此列表将显示与该设备兼容的可用驱动程序，以及与该设备属于同一类别的所有驱动程序。
第三步：选择点击此处
取消

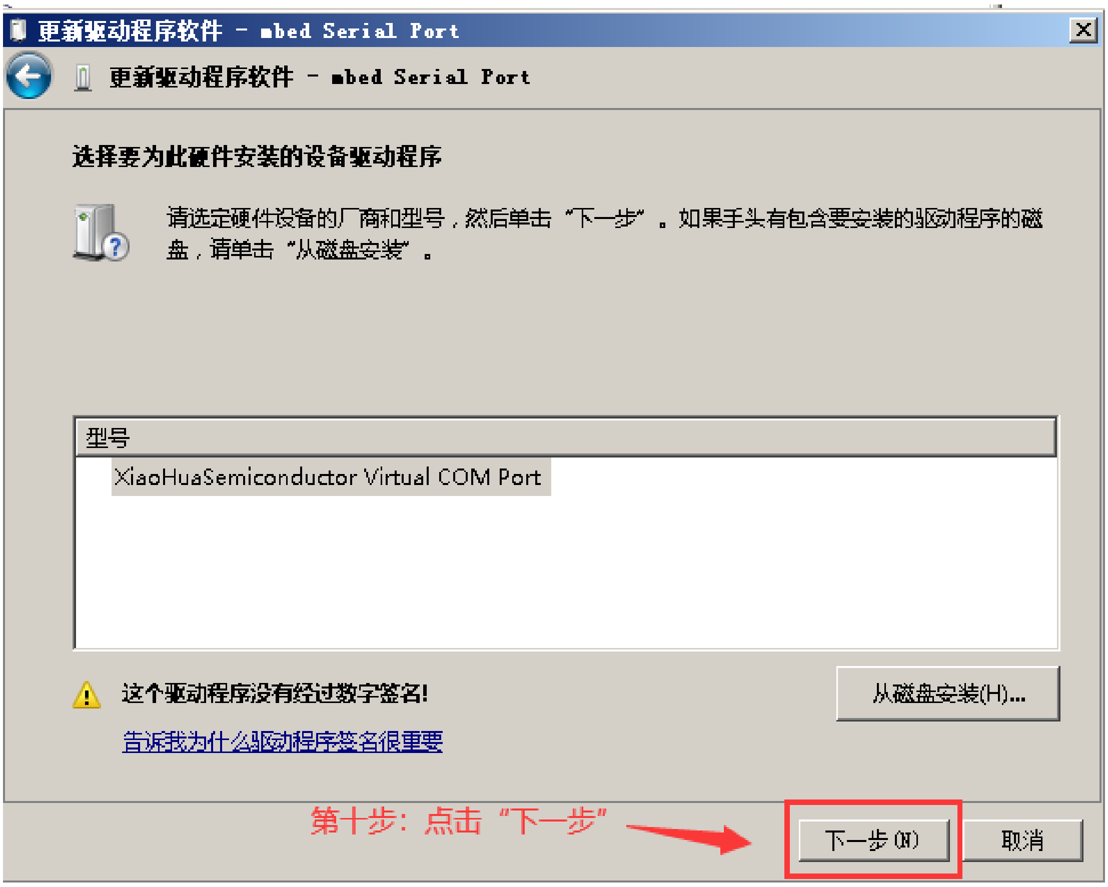

更新驱动程序软件 - mbed Serial Port
从以下列表选择设备的类型。
常见硬件类型(H):
显示所有设备
61883 设备类
AVC 设备
Bluetooth 无线电收发器
DVD/CD-ROM 驱动器
IDE ATA/ATAPI 控制器
IEEE 1284.4 兼容打印机
IEEE 1284.4 设备
IEEE 1394 和 SCSI 打印机
IEEE 1394 总线主控制器
Jungo Connectivity
Media Center Extender
第四步：直接选择“下一步”
下一步(N)
取消


更新驱动程序软件 - mbed Serial Port
选择要为此硬件安装的设备驱动程序
请选定硬件设备的厂商和型号，然后单击“下一步”。如果手头有包含要安装的驱动程序的磁盘，请单击“从磁盘安装”。
(正在检索所有设备的列表)
第五步：选择“从磁盘安装”
从磁盘安装(H)...
下一步(N)
取消

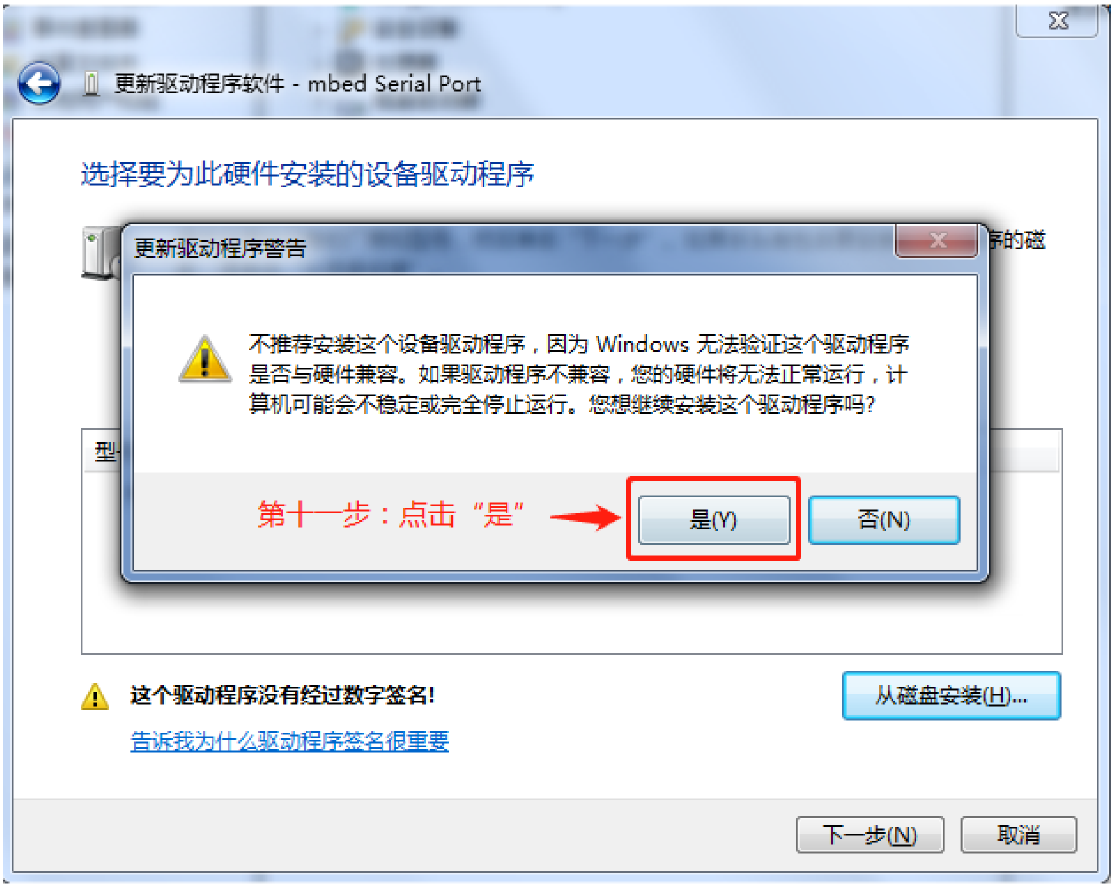

更新驱动程序软件 - mbed Serial Port
选择要为此硬件安装的设备驱动程序
请选定硬件设备的厂商和型号，然后单击“下一步”。如果手头有包含要安装的驱动程序的磁
从磁盘安装
插入制造商的安装盘，然后确定已在下面选定正确的驱动器。
确定
取消
第六步：点击“浏览”
制造商文件复制来源(C):
A:\
浏览(B)...
从磁盘安装(H)...
下一步(N)
取消


查找文件
第七步：选中由XHSC提供的驱动文件
查找范围(I):
vcom_driver_xhsc
名称
修改日期
类型
xhsccdc.inf
2022/1/27 9:45
安装信息
第八步：双击此文件
最近访问的位置
桌面
库
计算机
网络
文件名(N):
xhsccdc.inf
打开(O)
文件类型(T):
安装信息 (*.inf)
取消


更新驱动程序软件 - mbed Serial Port
选择要为此硬件安装的设备驱动程序
请选定硬件设备的厂商和型号，然后单击"下一步"。如果手头有包含要安装的驱动程序的磁盘，请单击"从磁盘安装"。
从磁盘安装
插入制造商的安装盘，然后确定已在下面选定正确的驱动器。
确定
取消
第九步：点击"确定"
制造商文件复制来源(C):
D:\vcom_driver_xhsc
浏览(B)...
厂商
%StdMf
(Generic
(Standar
(WPD 增
这个驱动程序已经过数字签名。
从磁盘安装(H)...
告诉我为什么驱动程序签名很重要
下一步(N)
取消

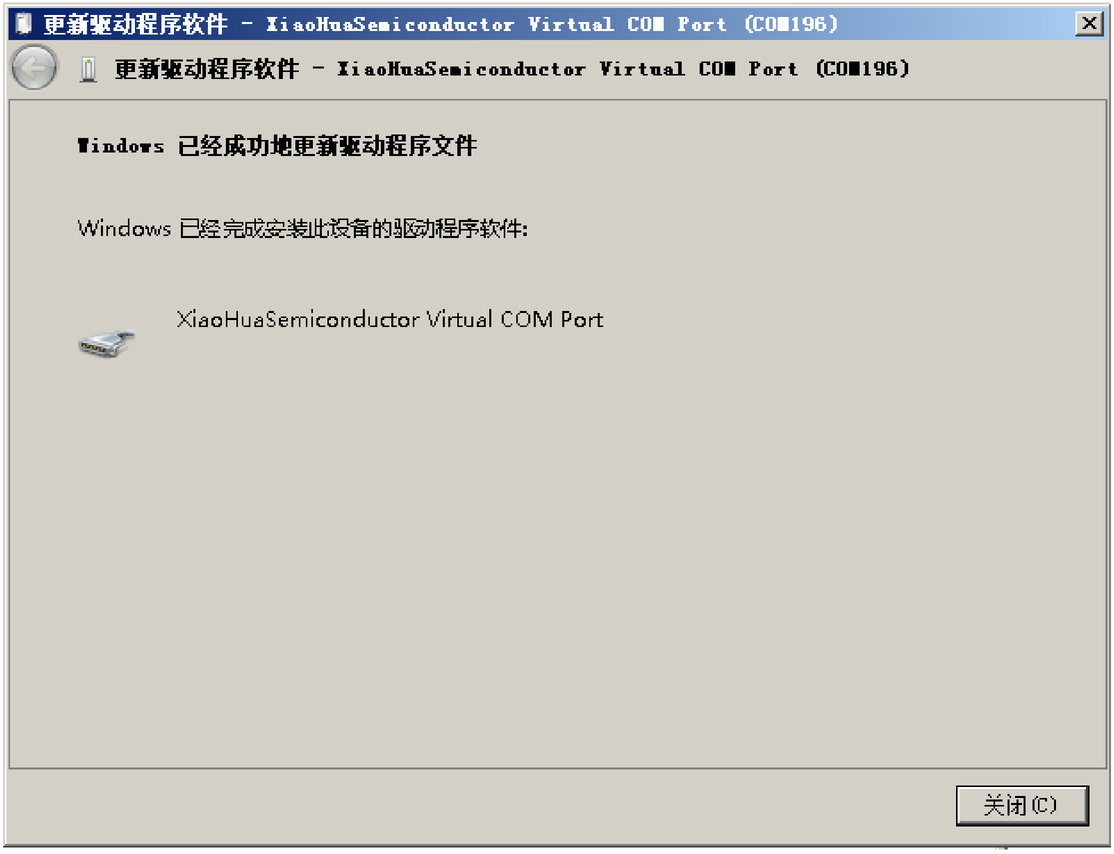


更新驱动程序软件 - mbed Serial Port
更新驱动程序软件 - mbed Serial Port
选择要为此硬件安装的设备驱动程序
请选定硬件设备的厂商和型号，然后单击 “下一步” 。如果手头有包含要安装的驱动程序的磁盘，请单击 “从磁盘安装” 。
型号
XiaoHuaSemiconductor Virtual COM Port
这个驱动程序没有经过数字签名!
告诉我为什么驱动程序签名很重要
从磁盘安装(H)...
第十步：点击 “下一步”
下一步(N)
取消



更新驱动程序软件 - mbed Serial Port
选择要为此硬件安装的设备驱动程序
更新驱动程序警告
不推荐安装这个设备驱动程序，因为 Windows 无法验证这个驱动程序是否与硬件兼容。如果驱动程序不兼容，您的硬件将无法正常运行，计算机可能会不稳定或完全停止运行。您想继续安装这个驱动程序吗?
第十一步：点击 “是”
是(Y)
否(N)
这个驱动程序没有经过数字签名!
告诉我为什么驱动程序签名很重要
从磁盘安装(H)...
下一步(N)
取消

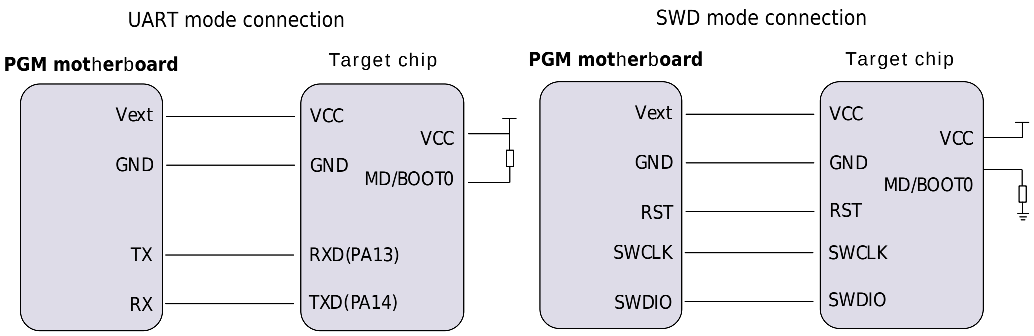
The driver starts to be installed. After a few seconds, the following screen is displayed, indicating that the installation is correct. 。



4.2 Program downloading

HC32L072_HC32L073_HC32F072 series MCU can be programmed through Xiaohua programmer.

The online programmer supports UART mode and SWD mode. The wiring method is shown in the figure below:



For the specific programming process, please go to the official website of Xiaohua Semiconductor https://www.xhsc.com.cn to find the corresponding chip model and refer to the Xiaohua programmer information for operation.

4.3 Low power mode program debugging

In the application, if the chip used has a low power mode and needs to enter the low power mode, the program will not be able to use the debugging function because the SWD function is disabled.

If this function is needed in the program, it is recommended to add a delay program of a few seconds at the beginning of the program during the debugging and development stage, or add an external IO control program to determine whether to execute the program, or add an external wake-up mechanism so that the SWD function can be used normally during the secondary debugging and development.

5 Development tool board code

5.1 Download and use of board code

The chips used in this series of EVBs support mainstream development environments such as IAR keil/ MDK; please go to the official website https://www.xhsc.com.cn. Select the category and model of the current chip in the development toolbar, and download the corresponding EVB board code.

5.2 Functional description of accompanying code

The board code functions used by this series of EVB boards include:

1) After power-on, the power indicator light is always on, and the LCD screen displays "HC:32";

2) USER buttons:

User button K1 triggers, D5-LED lights up;

User button K2 triggers, D6-LED lights up;

User button K3 triggers, D4-LED lights up;

User button K4 triggers, D3-LED lights up;

User button K5 triggers, D3/ D4/ D5/ D6-LEDs all light up.

3) EEPROM: MCU and BL24C02 exchange data through IIC protocol;

4) Audio: MCU and WM8731 communicate through IIS and IIC protocol, and connect headphones to output a sound.

Revision history

Version	Date	Revision Content
Rev1.0	2019/11/25	Initial version released.
Rev2.1	2020/11/13	Hardware version image changed; hardware corresponding silk screen changed; chip pin name changed (e.g.: mode→boot0); description text adjusted (e.g.: on-board debugging system→CMSIS DAP); development tool installation instructions deleted, etc. For details, see the document "Xiaohua Semiconductor MCU Development Environment Usage"; according to the hardware version number, the manual version number changed to Rev2.1.
Rev2.2	2022/07/15	Company Logo updated.
Rev2.3	2023/12/12	1) Adjust the content organization structure and update the template, modify some content descriptions and other details; 2) Add new board code.